

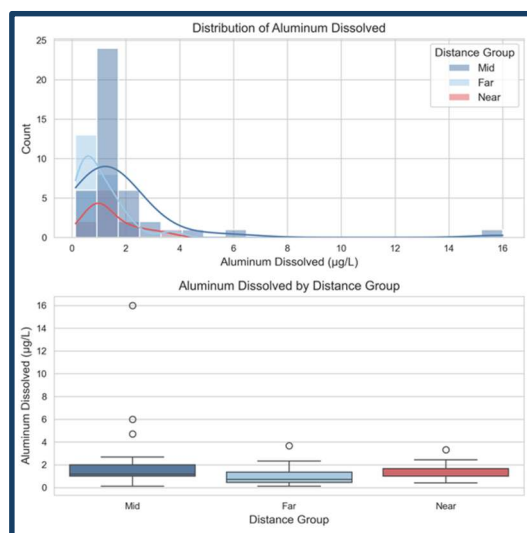
2.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to examine the structure, distribution, and quality of the dataset prior to formal statistical and machine learning analysis. The primary objective of EDA was to identify patterns, assess variability across parameters, and guide the selection of appropriate analytical methods, while ensuring that the dataset was suitable for multivariate analysis.

Initial visualizations focused on the distribution of key chemical parameters across distance groups. Boxplots were used extensively to assess central tendency, spread, and the presence of outliers for variables such as phosphorus, turbidity, pH, and selected trace metals (Topçu et al., 2026). These visualizations provided a consistent framework for comparing Near, Mid, and Far lake groups and were later retained in the Results section to support statistical findings. The use of boxplots allowed for clear identification of variability within and between groups without assuming normal distributions (Mcgill et al., 1978). An example of this approach is shown in Figure 2.4.1. These visualizations provided a consistent framework for comparing Near, Mid, and Far lake groups and were designed to evaluate differences in spread without assuming normal distributions.

Figure 2.4.1.

Distribution and boxplot of aluminum dissolved concentrations across distance groups, illustrating variability and outlier presence without assuming normality.



Time-based trends were explored using line plots of annual mean concentrations for selected variables, including phosphorus and turbidity (Figures 2.4.2 and 2.4.3). These plots were used to evaluate temporal consistency in the dataset and to identify any potential shifts or irregularities across sampling years (Dornelas et al., 2013). While temporal patterns were not the primary focus of this study, these visualizations ensured that no major inconsistencies or anomalies would bias subsequent analyses.

Figure 2.4.3.

Annual mean phosphorus total concentrations by distance group, used to assess temporal consistency and identify potential anomalies across sampling years.

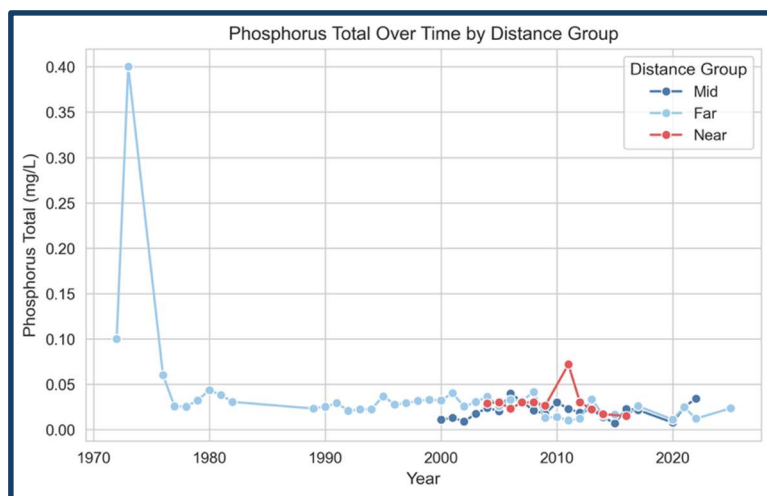
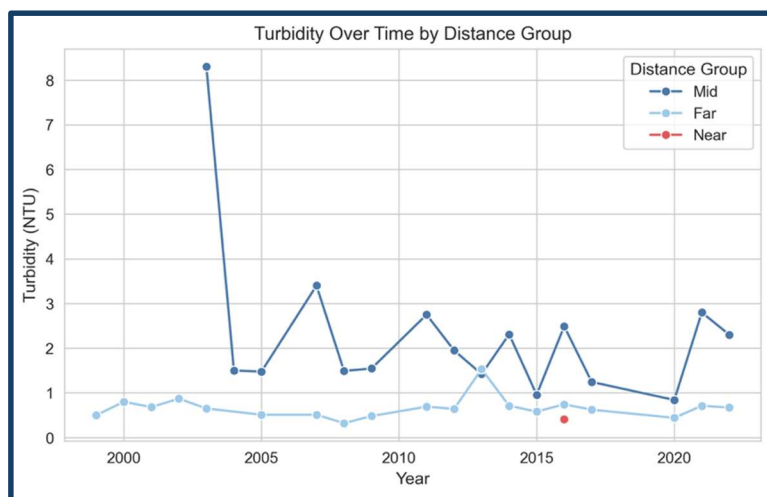


Figure 2.4.4.

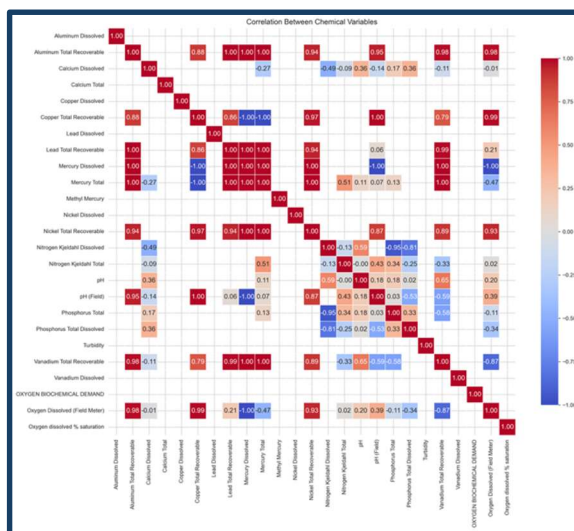
Annual mean turbidity levels by distance group, illustrating variability and potential fluctuations in water clarity over time.



To further examine relationships among variables, a correlation heatmap was generated using all standardized chemical parameters (Figure 2.4.4). This visualization provided insight into the degree of association between variables and helped identify potential multicollinearity, supporting the use of dimensionality reduction techniques such as Principal Component Analysis (Ivosev et al., 2008).

Figure 2.4.4.

Correlation matrix of water chemistry variables used in the analysis. Values represent Pearson correlation coefficients between standardized variables.



Spatial structure was examined using an interactive map of lake locations, with marker size representing sample count and color indicating distance group classification. This visualization provided geographic context for the dataset and confirmed the spatial distribution of sampling sites relative to the oil sands region. Additionally, summary tables were generated to assess the number of samples per lake and per distance group, ensuring adequate representation across categories.